Python Engineering - Reference Guide

Class 11 - Coordinate Transforms: 2D rotations

A simple guide to rotation matrices, matrix multiplication, and inverse rotations.

1. np.deg2rad()

What it does: converts degrees to radians.

What it receives: a scalar or array in degrees.

What it returns: the same scalar/array in radians.

```
theta = np.deg2rad(90.0)
# pi/2 approx 1.5708
```

2. np.cos() and np.sin()

What they do: compute cosine and sine element-wise.

What they receive: angles in radians.

What they return: values between -1 and 1.

3. 2D rotation matrix

```
R = [[ cos(theta), -sin(theta)],
     [ sin(theta),  cos(theta)]]
```

R has shape (2,2). Under the active convention it changes vector direction while preserving length.

4. The @ operator

What it does: matrix / linear-algebra multiplication.

What it receives: arrays with compatible dimensions.

What it returns: the corresponding linear product.

```
R.shape # (2,2)
v.shape # (2,)
R @ v  # (2,)
```

R * v would be element-wise multiplication/broadcasting; R @ v is the geometric operation we want.

5. np.linalg.norm()

What it does: computes vector magnitude.

What it receives: a numerical vector/array.

What it returns: a non-negative scalar.

```
np.linalg.norm(np.array([3.0,4.0]))
# 5.0
```

6. .T - transpose

What it does: swaps rows and columns.

What it receives: it is an ndarray property, not a function.

What it returns: a transposed view of the array.

7. Why is R.T the inverse?

A rotation matrix is orthogonal and satisfies:

```
R.T @ R = I
therefore R^-1 = R.T
```

8. np.allclose()

What it does: checks approximate equality using tolerances.

What it receives: two comparable values/arrays.

What it returns: True when all elements are sufficiently close.

9. Floating-point error

Real numbers have finite computer precision. Therefore cos(90 deg) may appear as 6.123e-17 rather than exact zero.

10. Active rotation vs frame transformation

EX19 uses an active rotation: the vector rotates while axes remain fixed. In a passive/frame transformation, the physical vector stays the same but its coordinates change when expressed in another frame. The matrices typically appear transposed/inverted between the two conventions.

Quick summary

Element	Input	Output / meaning
np.deg2rad(x)	degrees	radians
np.sin / np.cos	radians	sine / cosine
R @ v	compatible matrix/vector	rotated vector
np.linalg.norm(v)	vector	magnitude
R.T	matrix	transpose
np.allclose(a,b)	two arrays/values	approximate True/False